Republic of the Philippines
MINDANAO STATE UNIVERSITY – MAGUINDANAO
Dalican, Datu Odin Sinsuat, Maguindanao

COLLEGE OF ARTS & SCIENCES

LEARNING MODULE in ITE191 CAPSTONE PROJECT 1

BS Information Technology

Republic of the Philippines
MINDANAO STATE UNIVERSITY – MAGUINDANAO
Dalican, Datu Odin Sinsuat, Maguindanao

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ITE191

CAPSTONE PROJECT 1

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1st Sem. 2021-2022

Vision, Mission, and Core Values of the University

Vision

The premier innovative center of development in the region.

Mission

To train and enhance human capital through:

- Quality educational experiences;
- Relevant research and innovations and regional and national development; and
- Responsive community engagement and empowerment

Core Values

R- Responsiveness
I-inclusiveness
S- Sustainability
In- Innovativeness
G- Global Competence
In- Integrity
Collaboration

"It is in working together and collaborating with one another that we rise as one."

-Consuelo D. Samson, PhDLE
Vice Chancellor for Academic Affairs

Goal and Objectives of the College

GOAL

The **COLLEGE OF ARTS & SCIENCES** aims to produce quality graduates in Information and Communication Technology, Islamic Studies, Political Science and Peace and Development Studies.

OBJECTIVES

1. Deliver quality and modernized and modernized standard instruction in the fields of Arts & Sciences;
2. Equip students with intellectual, physical, social, psychological, and spiritual skills;
3. Produce globally competitive graduates;
4. Engage in research and extension services in different disciplines;
5. Strengthen linkages with Government, Private and Non-Government Organizations;
6. Ensure exclusive and equitable quality education that promote global learning opportunities;
7. Adapt to universal access for global degree of improvement in efficiency; and
8. Imbibe the concepts of peace, peacebuilding, diversity, inclusiveness, and respect in the curriculum as part of the pursued profession.

TABLE OF CONTENTS

LESSON 1
REVIEW OF THE INFORMATION TECHNOLOGY DOMAIN

Overview:

This lesson orients the students about capstone project and the domain of the Information Technology.

Learning Outcomes:

At the end of this lesson, the students can:

1. define what capstone project means;
2. identify the domain of Information Technology project;
3. differentiate capstone project and thesis; and
4. formulate capstone project proposed title.

Materials Needed:

- Computer
- Projector
- Presentation material

Duration: 1 hour and 30 minutes

Learning Content:

Capstone Project

A capstone is the top and last stone in a building. Similarly, a capstone course is the peak and last experience for students in a higher education program. Depending on the discipline and institution, capstone courses may take the form of group projects, senior seminars, undergraduate theses, or clinical experiences that integrate and synthesize what students have learned through the academic program.

Role of a Capstone Course

Capstone courses offer undergraduate students nearing graduation the opportunity to summarize, evaluate, and integrate some or all of their college experiences. Historically, capstone courses have been viewed as a "finishing touch" to provide students with the needed information or skills before graduation, hence the name "capstone." Recently, the prevailing perspective is that a capstone

course is an opportunity for students to demonstrate that they have met the goals established by their educational program.
https://www.slideshare.net/ledorsanecrab/capstone-primer-for-bsit-program-dbtc

Prerequisites for the Project

- Communications for IT – for formal articles/technical writing and presentation skills
- Systems Analysis and Design – for Software Development Life Cycle
- Applied General Statistics – for statistical process/treatment
- Professional Ethics – for intellectual property, and ethical and social implications of the project
- Software Engineering – for software development and paradigms
- Core Subjects (Programming Fundamentals, Data Structures, OOP, and DBMS)

Suggested Areas for Capstone Projects

1. Software Development
   - Software Customization
   - Information Systems Development for an actual client (with pilot testing)
   - Web Applications Development (with at least alpha testing on live servers)
   - Mobile Computing Systems
2. Multimedia Systems
   - Game Development
   - e-Learning Systems
   - Interactive Systems
   - Information Kiosks
3. Network Design and Implementation and Server Farm Configuration and Management
4. IT Management
   - IT Strategic Plan for sufficiently complex enterprises
   - IT Security Analysis, Planning and Implementation

Learning Activities:

Activity 1:

Research the difference between Capstone Project and Thesis.

Activity 2:

Given the list of titles below, identify whether it is a research or capstone, the domain where it belongs.

Capstone Projects and Thesis Titles for Information Technology

______1. Workflow Management System for MNC
______2. Secured Merchant Payment using Biometric Transaction
______3. Online Personal Counselling
______4. E-Visa Processing & Follow Up System
______5. Loss Prevention System in Stock Market Trading
______6. E Commerce for Online Medicine Shopping
______7. Online Transaction Fraud Detection using Backlogging on E-Commerce Website
______8. Employee Performance Evaluation & Appraisal Calculation using Data Mining
______9. Online Secondhand Book Buying & Selling Portal
______10. College E Print Service Management
______11. Internet based Discussion Forum
______12. Increase Productivity Using Quality Management System
______13. Diabetes Prediction Using Data Mining
______14. On Road Vehicle Breakdown Assistance Finder
______15. Online Faculty Staff Directory for Multi University
______16. Customer Targeted E-Commerce
______17. Data Mining for Sales Prediction in Tourism Industry
______18. Hotel Recommendation System Based on Hybrid Recommendation Model
______19. Online Health Shopping Portal With Product Recommendation
______20. College Forums with Alumni Based on Content Filtering
______21. Advanced Intelligent Tourist Guide
______22. Industrial Visit Planning & Booking System
______23. Intelligent Tourist Guide
______24. Online Pizza Ordering System
______25. Personality Prediction System Through CV Analysis
______26. Secure Backup Software System
______27. TV Show Popularity Analysis Using Data Mining
______28. Twitter Trend Analysis Using Latent Dirichlet Allocation
______29. Secure E Learning Using Data Mining Techniques
______30. Price Negotiator Ecommerce ChatBot System
______31. Predicting User Behavior Through Sessions Web Mining
______32. Online Book Recommendation Using Collaborative Filtering
______33. Monitoring Suspicious Discussions On Online Forums Php
______34. Fake Product Review Monitoring & Removal For Genuine Ratings Php
______35. Detecting E Banking Phishing Using Associative Classification
______36. A Commodity Search System For Online Shopping Using Web Mining
______37. Detecting Phishing Websites Using Machine Learning
______38. Secure Electronic Fund Transfer Over Internet Using DES
______39. Sports Events Management Platform for Colleges
______40. Secure Online Auction System
______41. Filtering political sentiment in social media from textual information

_______42. Evaluation of Academic Performance of Students with Fuzzy Logic
_______43. Cooking Recipe Rating Based On Sentiment Analysis
_______44. Student Information Chatbot Project
_______45. Website Evaluation Using Opinion Mining
_______46. Social Media Community Using Optimized Clustering Algorithm
_______47. Preventing Phishing Attack On Voting System Using Visual Cryptography
_______48. Search Engine Using Web Annotation
_______49. Secure File Storage On Cloud Using Hybrid Cryptography
_______50. E Banking Log System

https://www.inettutor.com/source-code/capstone-projects-and-thesis-titles-for-information-technology/

Learning Evaluation:

1. Submit your capstone project proposed title.

References:

URL

Barcenas, R., Teruel, A. (2012). *A Handbook for the Capstone Program of the BSIT Department.* https://www.slideshare.net/ledorsanecrab/capstone-primer-for-bsit-program-dbtc, Date Accessed Nov. 13, 2013

https://www.inettutor.com/source-code/capstone-projects-and-thesis-titles-for-information-technology/; Date accessed Aug. 17, 2021

LESSON 2
OVERVIEW OF THE RESEARCH PROCESS

Overview:

This lesson explains the process of conceptualizing, gathering information, analyzing and writing research or project.

Learning Outcomes:

At the end of this lesson, the students can:

1. outline the plan to conduct the capstone project;
2. discuss the research process used in the conduct of capstone project; and
3. compose a process of research with a given capstone project title.

Materials Needed:

- Computer
- Projector
- Presentation material

Duration: 1 hour and 30 minutes

Learning Content:

Capstone Process
The Capstone process is comprised of two phases:

Phase 1
The first phase has been designed to assist students in developing and creating a formal Capstone Project Proposal. The Capstone Mentor will review the student's Project Proposal and make recommendations before granting approval to proceed.

Phase 2
The second phase has been designed to assist the student in implementing their approved formal Capstone Project Proposal toward completion of their project. As the student progresses toward completing their project, their Mentor will likely wish to have the opportunity to review and critique the Capstone Project activities and the formal documentation on a section by section basis so that any significant

changes or amendments that are suggested can be integrated into the final presentation well before the submittal of a final paper.

Capstone Project | Aspen University, 2013

Research Process

Research is a process. It proceeds from one to the next in a progression until its completion. At some point in the process, researchers need to go back to rethink, re-evaluate or repeat.

The simplified and general steps are as follows:

1. Identify the question or problem.

This can be a problem that needs to be solved or some piece of information that is missing about a particular topic. Answering this question will be the focus of the research study. (Offord Centre for Child Studies, 2017)

2. Review the existing literature

The researcher must learn more about the topic they are investigating. This not only provide important background information about the issue but also tells what other studies have already been concluded, how it was designed and what those studies found.

3. Clarify the problem

Original research question may be too broad to examine that knowledge from literature review can be used to narrow the focus or scope of the study.

4. Clearly define the terms and concepts

Terms and concepts are used in the purpose statement or description of the study. These items need to be specifically defined as they apply to the study.
By defining terms or concepts, the scope of the study is more manageable making it easier to collect necessary data for the data.

5. Define the population

Population is the group to involve in the study. Defining the population assists the researcher in:

a. Narrowing the scope of the study
b. Focusing on within the study
c. Identifying the group that the results will apply to at the conclusion of the study.

6. Select methods of data collection

This involves planning specifics of a study protocol, such as who will participate, exactly what type of data will be collected, and how, when, and where the data will be gathered.

7. Develop the instrumentation plan

Instrumentation plan serves as the road map for the entire study, specifying who will participate in the study, how, when, and where data will be collected and the content of the program.

The instrumentation plan specifies all the steps that must be completed for the study.

8. Collect data

The collection of data is a critical step in providing the information needed to answer the research question. Data can be collected in the form of words on a survey, with a questionnaire, through observations or from the literature.

9. Analyze data

Analysis of data plays an important role in the achievement of research aim and objectives. Once data has been collected, it must be analyzed in order to answer the original research question. Appropriate methods must be chosen in order to come up with a valid answer.

10. Write your paper

Organize all information collected, get ideas on paper. This will help determine the form your final paper will take.

11. Cite sources properly

Give credit where credit is due.

Citing or documenting the sources gives proper credit to the authors of the materials used and it allows those who are reading duplicates your research and locate the sources of your references.

12. Conclude your research

Conclusions relate to the level of achievement of research aims and objectives. It justifies that the aims and objectives have been achieved.

13. Proofread

Proofread the paper created – reading through the text, and check for any errors in spelling, grammar, and punctuation. Making sure the sources used are cited properly and making sure the message get across to the reader has been clearly and thoroughly stated.

14. Share results

Share the results of the research through publication, through sending or submission to persons concerned or stakeholders such as policy makers, government officials, company or business executives, through presentation to an audience or public.

The results are shared so that other researchers can learn from the study or can be used to influence policies and programs.

Learning Activities:

Activity 1:

Research and discuss different research process used in the conduct of capstone project.

Activity 2:

Given a title of a capstone project. Suggest a research process to conduct the said research.

Learning Evaluation:

1. Outline the conduct of the capstone project you have chosen.

References:

Books

- Almeida, A., et al (2016). *Research Fundamentals: From Concept to Output. A Guide for Researchers* & *Thesis Writers.* Adriana Printing Co., Inc.

LESSON 3
IDENTIFYING THE RESEARCH TOPIC OR ORGANIZATIONAL ISSUE

Overview:

This lesson discusses a way to find a working title for a research.

Learning Outcomes:

At the end of this lesson, the students can:

1. find a capstone project topic; and
2. write a working title that describes what the study is all about.

Materials Needed:

- Computer
- Projector
- Presentation Material

Duration: 1 hour and 30 minutes

Learning Content:

When finding the right research paper topics think of several subjects that interest you. Try writing down these subjects and narrow down topics you are most interested. (Collins, 2014)

Once you have your chosen topic, check for references in the internet, library, books, articles, encyclopedia. The more research materials the easier to about write your research paper.

Research Problem

A research problem is considered a professional situation in need of improvement, change or solution.

Elements of Research Problem

1. Aim or purpose of the problem for investigation. This answers the question “Why?”

- Why is there an investigation, inquiry or study?

2. The subject matter or topic to be investigated. This answers the question "What?"

- What is to be investigated or studied?

3. The place or locale where the research is to be conducted. This answers the question "Where?"

- Where is the study to be conducted?

4. The period or time of the study during which the data are to be gathered. This answers the question "When?"

- When is the study to be carried out?

5. Population or universe from whom the data are to be collected. This answers the question "Who?" "From whom?"

- Who are the respondents?
- From whom are the data to be gathered?

Characteristic of Research Problem S.M.A.R.T.

Specific: The problem should be specifically stated.
Measurable: It is easy to measure by using research instruments, apparatus, or equipment.
Achievable: Solutions to a research problem are achievable or feasible.
Realistic: Real results are attained because they are gathered scientifically and not manipulated or maneuvered.
Time-bound: Time frame is required in every activity because the shorter completion of the activity, the better.

Sources of Research Problem

- Specialization of the researcher
- Current and Past Researches
- Recommendations from theses, dissertations, and research journals
- Original and creative ideas of the researcher based on the problems met in the locality and country

Criteria of a Good Research Problem

- Interesting
- Innovative
- Cost-effective
- Relevant to the needs and problems of the people
- Relevant to government's thrusts

- Measurable and time-bound

Concept Mapping

A structured approach that groups can use to map out organize their ideas on any topic is called concept mapping. This can be used by research teams to help them clarify and map out the key research issues in an area, to help them operationalize the programs on interventions on the outcome measures for the study. The concept mapping method isn't the only method around that might help researchers formulate good research problems and methods and projects. Some of the methods that might be included in the toolkit for research formulation might be: brainstorming, brain writing, nominal group techniques, focus group, etc. (Trochim, 2006)

According to Katsumoto, 2005 (cited in de Belen, 2015) concept mapping is a general method with which you clarify and describe people's ideas about some topic in a graphic form. By mapping out concepts in pictorial form you can get a better understanding of a relationships among them. Concept mapping encourages the participants rather than on the planner or evaluator.

Research Title

The research title summarizes the main idea or ideas of your study. A good title contains the fewest possible words needed to adequately describe the content and/or purpose of your research paper.

According to Maxine Hairston and Michael Keene, a good title has several functions:

1. It predicts content
2. It catches reader's attention
3. It reflects tone or slant of the piece of writing
4. It contains keywords that will make it easy to access by a computer search

How to Write a Working "Research Title"

1. Ask these questions and make note of the answers:
a. What is the paper about?
b. What techniques/ design will be used?
c. Who/what will be studied?
2. Use the answers to list keywords
3. Create a sentence that includes the keywords listed

4. Delete all unnecessary/ repetitive words
5. Delete non-essential information and reword the title.

Characteristics of Title

1. Indicate accurately the subject and scope of the study
2. Rarely use abbreviations or acronyms unless they are commonly known
3. Use words that create a positive impression and stimulate reader interest
4. Use current nomenclature (terminology) from the field of study
5. Reveal how the paper will be organized
6. Is limited to 5 to 15 substantive words
7. Does not include redundant phrasing, such as “A Study of”, “An Analysis of” or similar constructions
8. Takes the form of a question or declarative statement
9. If quote is used as part of the Title, the source is cited (usually using an asterisk and footnote)
10. Use correct grammar and capitalization. The first letter of the title and all nouns, pronouns, verbs, adjectives and adverbs that appear in the title are capitalized.

Learning Activities:

Activity 1:

Read the sample document below…

> People nowadays are living in an information age dependent upon digital Information. Digital information is electronic information, the result of computer processing. Every type of job relies upon getting information, using it, managing it, and relaying information to others. Computers enable the efficient processing and storage of information.
>
> A grading system plays a key role in the management system of any school. But, such systems do not often relate expectations, outcomes, and performance. As each student desires to achieve a good score for each assignment, exam, project and/or report, the whole process adds heavy workload for teachers in order to make their evaluation fair, comprehensive, and accurate. From the faculty perspective, these are necessary to avoid disagreement from students and parents. A computerized grading system is

a highly desirable addition to the educational tool-kit, particularly when it can provide less effort and a more effective and timely outcome.

Grading systems are designed to provide incentives for achievement and assist in identifying problem areas of a student. It is the most commonly used means of analyzing student performance, talents and skills. Students' grades are vital information needed in advancing to the next grade/year level and its accuracy is very important.

Many teachers feel that the time they take in recording and computing for the grades of their students is time that could be better spent elsewhere, like preparing lessons, researching or meeting with their students. With the advent of computer technology, more and more schools are taking advantage of a variety of grading systems available both off-line and online that can help save time on administrative tasks and give teachers more time to attend to other important functions. However, a greater majority, especially small schools, government schools, and schools in remote areas, still utilize the manual method of recording and computing for the grades of the students.

The proponents of this study wish to reduce the workload of teachers by eliminating the need for manual computation and recording of each grade. The common problems encountered in manual recording and computations are error- and file-handling, and redundancy. As the workload gradually increases with growing amounts of grades and student lists that need to be attended, it becomes tedious on the part of the teacher to proficiently manage them in time for documentation and file submission to higher education authorities. As such, this paper aims to produce a workable computerized grading system that will address these issues.

1. Answer the following questions:
    a. What is the problem?
    b. Why is it a problem?
    c. Who is/are involved in the problem?
    d. Where is the problem to be conducted?
    e. When will the study be conducted?

2. Write a possible title for the document above.

Learning Evaluation:

1. Submit 3 research topics for your capstone project following this format:

Research Topic or Title	
What is the Problem	
Why is it a Problem?	
Where is the Problem?	
Who is/are involved in the problem?	
When is the conduct of the study?	

References:

Books

- Almeida, A., et al (2016). *Research Fundamentals From Concept to Output. A Guide for Researchers & Thesis Writers.* Adriana Printing Co., Inc.

- Del Rosario-Garcia, M., et al (2017). *Practical Research 1 Basics of Qualitative Research*. Fastbooks Educational Supply Inc.

LESSON 4
PREPARING AND WRITING A LITERATURE REVIEW

Overview:

Related literature and studies consist of written documents and works. It systematically presents facts, theories, constructs, concepts, variables, and measures related to the study as culled from documents.

Learning Outcomes:

At the end of this lesson, the students can:

1. organize the review of related literature and studies;
2. demonstrate skill in acknowledging sources of documentation or citation of references; and
3. construct literatures related to the research topic.

Materials Needed:

- Computer
- Projector
- Presentation material

Duration: 1 hour and 30 minutes

Learning Content:

Organizing the Review of Related Literature and Studies

Organized related literature and studies inform the reader of what is known and conflicting area. The word review means that the research goes over the materials, books. Journal articles, theses, dissertations and internet presentations. The purpose is to determine what has been written about the problem. The word related means similar especially on the problem dimension.

The purpose of the review is to inform the reader about what already is known, what is not known or research blank spots (unexplored areas) and blind spots (conflicting areas) in the literature.

How to Select and Review the Literature

1. Finding Information
    - You should know **what** information to look for and **where** to look for the information
      The **what** information to look must be answered by your topic, research questions and statement of the problem which are presumed that you already have at this point in time.
2. Evaluating Content

- Evaluate the quality and scholarliness of a source. Make sure you only include credible, scholarly sources.

Guidelines:

Authority	What are the author's credential? Can you identify their institutional affiliations? What is the author's expertise on the subject?
Currency	When was the source published? Last 5 years? Is it outdated? Does it meet the time needs for your topic?
Documentation	Does the author cite credible, authoritative sources? Is there evidence of scholarly research? Do they properly cite their sources?
Intended Audience	Who is the intended audience? Scholars? Researcher's? General audience?
Objective / Purpose	What is the author's goal in writing it? To entertain? To inform? To influence? How objective is the source?
Relevancy	Is it relevant to your topic? Does it provide any new information about your topic?

(Eastern University, 2017)

- If you are using the internet, you have to be mindful that not all materials that you find are credible and legit. Be cautions and critical about the materials found especially in many social networking sites and bogus news sites.

3. Recording the Information
    - Take down notes and use the method that works best for you.
        i. The Outline Method
        ii. The Mapping Method
        iii. The Charting Method
        iv. The Sentence Method
4. Synthesizing Content
    - Reflect on the research you have, what have you learned, how the information fits into your topic, and plan what is the best way to present your findings.
    - Organize research by topics.
    - Consider points from each topic you want to address. Start thinking about areas you will discuss in your review and what pieces of research you will use to support your conclusions.
    - Try summarizing the main points in one or two sentences.
    - Draft an outline of your literature review.
5. Write the Review
    - Present relevant findings and issues from research articles in literature review
    - Decide on organization pattern into a coherent presentation: chronological, categorical/topic, general to specific, known to unknown, etc.
    - Follow the models of the research articles of previous research as closely as possible
    - Synthesize information from studies into meaningful presentation. Use only topics that help advance the necessary definitions,

context, explanation and rationalization for variable, gaps and the argument/rationale for the study

- The rationale. The explanation that leads to the research questions. What research questions can you derive from your examination of the previous literature?

Related Literature

Related literature is also called conceptual literature. The gathering of literature related to the research clarifies the different variables being studied. The related literature removes any vagueness in the concepts central to the study. As the related literature and studies are gathered, different variables of the study are clarified, the limitation of the study is determined. The indicators per variable are clearly based on read articles from different reference books, manuals and other materials. (Cristobal & Cristobal, 2013)

By organizing, integrating and evaluating such materials, the author of a review article considers the progress of current research towards clarifying a problem.

The different variables used in the study are the focus of the literature search. The title, statement of the problem, score and limitation, including the paradigm of the study are the elements that give the research an idea of the relevance of the thesis to his/her own investigation.

The number of materials to be gathered for the literature review depends upon the researcher's judgement. The review of related literature is considered sufficient if the researcher believes that important concepts and variables have been adequately explained and established.

Related Studies

Determining whether the research is objective and empirically-based includes surveying previous studies that involve similar variables (Cristobal & Cristobal, 2013). It is important to note that even if a previous research used the same variables as his/hers, the twos studies may vary in the limitation in terms of the sub-variables investigated or in terms of focus and purpose.

An adequate review of related studies is needed. This serves as the basis of the analysis of results because it allows the researcher to compare and contrast his/her findings with those of previous studies. The result of a study are verified by similar findings or negated by different findings of previous researchers.

The studies are in the form of theses, dissertations or journal articles. These are collectively called research literature.

The collected studies are related to the present study when the author uses the same variables, sub-variables, concepts or constructs, and have the same subject or topic of the study.

In the review of related study, the following are indicated: 1. The name of the author, the date and the setting the study was conducted; 2. The title 3. The salient findings.

Documentation style

Throughout the research, the published researches of other researchers are cited to credit those who prepared the foundation for your work. To present someone else's ideas or work as your own is to commit plagiarism. (Gravetter & Forzano, 2006)

The American Psychological Association (APA) is the popular style commonly used by researchers in documentation or reference citation. The APA format is also called author-date method or parenthetical documentation or in-text citation.

Documentation refers to the acknowledgement of sources. These sources are cited as direct quotations or as indirect quotations (paraphrases).

Paraphrasing

Paraphrasing is using or putting in one's own words to restate the author's ideas and acknowledging the source to give credit to the original author (de Belen, 2015)

Paraphrasing

- is used when we want to express someone else's idea in our own words.
- Uses different words to express the same idea
- Is rewriting the next in order to simplify focusing not only on what is said but also on how it is said.
- Makes the understanding of the source text less difficult by breaking down the information into manageable units

Techniques

- Replace a word with a synonym
- Paraphrasing can be longer than the original. Concentrate on the meaning not on the words.
- Verbs and adjectives have counterparts that are interchangeable with the author's original words.
- Not all synonyms have exact meaning
- Avoid using abstract words – they come as weak words
- Keep important words (such as scientific writing) and change only the sentence structure.

Learning Activities:

Activity 1:

1. Find 1 literature related to this title "Computerized Grading System for Metropolitan Academy of Manila".
2. Answer the following questions:
   a. What is the theme/topic of the article?
   b. Who are the authors mentioned in the article?
   c. What are the information cited by the author(s)?

d. Write the reference of the literature/article based from the source following the APA format.

Learning Evaluation:

Direction: Answer what is being asked:

1. Find 10 literatures (local and international) related to your research topic.
2. Cite what have been said by the author/s.
3. Write the reference/ source of the literatures obtained.

References:

Books

- Almeida, A., et al (2016). *Research Fundamentals From Concept to Output. A Guide for Researchers & Thesis Writers.* Adriana Printing Co., Inc.
- Del Rosario-Garcia, M., et al (2017). *Practical Research 1 Basics of Qualitative Research*. Fastbooks Educational Supply Inc.

LESSON 5
PRESENTING PROJECT PROPOSAL: INTRODUCTION

Overview:

The main purpose of the introduction is to give a description of the research that will address a problem. This lesson will discuss the background of the research, purpose of the research, the research questions to be addressed and the significance of the research and scope of research.

Learning Outcomes:

At the end of this lesson, the students can:

1. write a research that describes what the study is all about;
2. write a background of the study;
3. compose statements and objectives of the research;
4. outline the scope and limitations of the research study; and
5. identify the significance of the study.

Materials Needed:

- Computer
- Projector
- Presentation Slide

Duration: 1 hour and 30 minutes

Learning Content:

Background of the Research

Project Context

1. The proponent should introduce the presentation of the problem, that is, what is the problem is all about. The proponent should describe the existing and prevailing problem situation based on his or her experience. This scope may be global, national, or regional and local.
2. The proponent should give strong justification for selecting such research problem in his/her capacity as a researcher. Being part of the organization or systems and the desire and concern to improve the systems.
3. The researcher state a sentence or two that would show the link and relationship of the rationale of the study to the proposed research problem.

Purpose and Description of the Project

- What is the function of your project?
- What is good in your project?
- What makes your project unique, innovative, and relevant?

Statement of the Problem

A problem statement is a brief piece of writing that usually comes at the beginning of a report or proposal to explain the problem or issue the document is addressing to the reader.

Objectives of the Project

1. Start with the **General Objective** which is very parallel to the project title.
2. Explode the general objective into **Specific Objectives** that will help realize the proposed study.
3. Objectives should be **SMART**

Scope and Limitation of the Project

- Think the project scope as a **box**. High-level scope defines the sides of the box and separates what is relevant to your project from what is irrelevant.
- The **scope** refers to the work that needs to be accomplished to deliver a product, service, or result with the specified features and functions.
- The **scope** explains the nature, coverage, and time frame of the study
- The **limitation**, on the other hand, explains all that are **NOT** included in your project.
- In other words, the **scope of the project** gives an overview all the **deliverables (i.e. the things that your project gives/delivers)**, and the **tools and technologies** used that will be used in the project development while the **limitations of the project** are the **boundaries** of the project **(i.e. areas/things that are out of scope)**.

Significance of the Study

The significance of the study is a brief declaration that such study is very important and necessary.

- Identify the beneficiary or beneficiaries
- Describe the “benefits or benefits” that will be derived from the research of the research or study.
  - if you have many beneficiaries, you can use the following format:

- 1st paragraph – why is it important to conduct this study
- 2nd paragraph – the main beneficiary – either a group or an organization
- 3rd paragraph – the secondary beneficiary
- 4th paragraph – the importance of the study to the proponents or researchers
- 5th paragraph – the importance to the future researchers

Learning Activities:

Activity 1:

Given a research topic on "Computerization of the Grading System in MSU Maguindanao", write an Introduction that answers the following questions:

a. Cite what is the problem?
b. Why is it a problem?
c. How should it be solved?
d. Why it should be solved?
e. What is the purpose of the study?
f. What are the scope and limitation of the study?
g. Who will benefit on the study?

Learning Evaluation:

1. Write an introduction (inclusive of the Background, statement of the problem, objectives of the study, scope and limitations and significance of the study) of the research topic of your choice.

References:

Books

- Almeida, A., et al (2016). *Research Fundamentals From Concept to Output. A Guide for Researchers & Thesis Writers.* Adriana Printing Co., Inc.
- Del Rosario-Garcia, M., et al (2017). *Practical Research 1 Basics of Qualitative Research*. Fastbooks Educational Supply Inc.

LESSON 6

QUANTITATIVE METHODS AND APPROACHES

Overview:

This lesson will lead you to understand the various aspects of the research process such as planning the structure of investigation to obtain answer to the research problem or questions.

Learning Outcome:

At the end of this lesson, the students can:

1. identify the research method appropriate to a research and justify their choice.

Materials Needed:

- Computer
- Projector
- Presentation Slide

Duration: 1 hour and 30 minutes

Learning Content:

Research Design

After the research topic has been finalized, the researcher has to plan the details of what design to use, what type of data will provide answers to the problems of the study, and how the data will be gathered, presented, analyzed and interpreted.

The research design guides the researcher in planning the following aspects or procedures of research:

- Identifying the population of the study
- Decision on whether to take the whole population or just select a sample
- How the sample of the study will be selected
- Ethics in the selection of samples and data gathering
- Choice of method in data collection
- Considerations in the use of questionnaires
- How interview will be conducted

Research Designs based on the Number of Contacts

1. Cross Sectional studies
   - Commonly used in social sciences
   - Aim to find out the prevalence of a phenomenon, situation, problem, attitude or issue, by taking a cross section of the population
2. Before and After (or Pre-Test / Post-Test) Design
   - Can measure change in a situation, phenomenon, issue, problem or attitude.
   - The change is measured by comparing the difference in the phenomenon or variable before and after the intervention.
3. Longitudinal study design
   - Useful to determine the pattern of extent of a change in a phenomenon, situation, problem or attitude in relation to time.
   - The study population is visited an number of times at regular intervals.

Research Designs based on the Reference Period

The reference period refers to the timeframe in which a study is exploring a phenomenon, situation, event or problem and may be categorized as retrospective, prospective and retrospective-prospective.

- Retrospective study design is used to investigate a phenomenon, situation, problem or issue that has happened in the past. The study may be conducted either on the basis of the data available for that period or on the basis or respondent's recall of the situation.
- Prospective of the study design attempts to establish the outcome of an event or what is likely to happen, such as likely prevalence of a phenomenon, situation, problem, attitude or outcome in the future. Experimentation is an example.
- Retrospective-prospective study design applies to a study wherein available data are analyzed and used as bases for future projections. Trend is an example.

Research Designs based on the Nature of Investigation

- The Experimental design has an assumption of a cause-and-effect relationship. The researcher introduces the intervention that is assumed to be the cause of change and waits until it has produced the change.
- In the Non-experimental design, the researcher observes a phenomenon and attempts to stablish what causes it. In this instance,

the research starts from the effect or outcome and attempts to determine causation.

- A semi-experimental or quasi-experimental has the properties of both experimental and non-experimental studies that part of the study may be experimental and the other part non-experimental.

Learning Activity:

Activity 1: Identify which research title is applicable to quantitative method and approach.

1. Do Advertisements Need To Be Censored Strictly?
2. Effect Of E-Marketing On Young Entrepreneurs.
3. Glass Ceiling: Is It Durable And Eco-Friendly?
4. Improving The Scores Of Paragraph Writing In A Fourth Grade Language Art Class Using State Test Release Items And Scoring Rubrics As Instructional Material.
5. Increasing Kindergarten Verbal Participation Rates Using Story Elaboration Techniques
6. Overcoming Adversity: Death Of A Loved One.
7. Students' Perception On School Environment
8. The Impact Of Using Word Walls In Teaching Fifth Graders Content Vocabulary In Science.
9. Using Competitive Flashcard Games To Improve Memory For Math Facts In Special Education.
10. Using Online Reading Sources In A Grade Ten History Class To Advance Students' Use Of Multiple Perspectives.

Learning Evaluation:

1. Given a research topic "Computerization of the Grading System in MSU Maguindanao", how will you make it a quantitative research?

References:

Books

- Almeida, A., et al (2016). *Research Fundamentals From Concept to Output. A Guide for Researchers & Thesis Writers.* Adriana Printing Co., Inc.

- Del Rosario-Garcia, M., et al (2017). *Practical Research 1 Basics of Qualitative Research*. Fastbooks Educational Supply Inc.

LESSON 7
QUALITATIVE METHODS AND APPROACHES

Overview:

This lesson will lead you to understand the various aspects of the research process such as planning the structure of investigation to obtain answer to the research problem or questions.

Learning Outcomes:

At the end of this lesson, the students can:

1. identify the research method appropriate to a research and justify their choice.

Materials Needed:

- Computer
- Projector
- Presentation Slide

Duration: 1 hour and 30 minutes

Learning Content:

Research Designs

1. Case Study
    - a very useful design when exploring an area where little is known or where you want to have a holistic understanding of the situation, phenomenon, episode, site, group or community. The design is relevant if the focus of the study is on extensively exploring and understanding rather than confirming and quantifying. (Kumar, 2011)
2. Grounded Theory
    - A process for developing empirical theory form qualitative research that consists of a set of tasks and underlying principles through which theory can be built up through careful observation of the social world.
3. Phenomenology
    - This is a qualitative research design which studies all possible appearance in human experience using empirical method (i.e. asking, observing, analyzing data, etc.) to make empirically grounded statement that can be generalized.
4. Ethnography

- This is a research process which deals with the scientific description of individual cultures involving the origins, development and characteristic of human kind, including social customs, belief and cultural development (Weirsma and Jurs, 2009)

5. Mixed Methods
    - Refers to the research design that uses both quantitative and qualitative data to answer a particular question or sets of questions. Words, pictures and narratives can be used to add meaning to numbers.

Learning Activities:

Activity 1: Identify which research title is applicable to qualitative method and approach.

1. Do advertisements need to be censored strictly?
2. Effect of e-marketing on young entrepreneurs.
3. Glass ceiling: Is it durable and eco-friendly?
4. Improving The Scores Of Paragraph Writing In A Fourth Grade Language Art Class Using State Test Release Items And Scoring Rubrics As Instructional Material.
5. Increasing Kindergarten Verbal Participation Rates Using Story Elaboration Techniques
6. Overcoming Adversity: Death of a loved one.
7. Students' Perception on School Environment
8. The Impact Of Using Word Walls In Teaching Fifth Graders Content Vocabulary In Science.
9. Using Competitive Flashcard Games To Improve Memory For Math Facts In Special Education.
10. Using Online Reading Sources In A Grade Ten History Class To Advance Students' Use Of Multiple Perspectives.

Learning Evaluation:

1. Given a research topic "Computerization of the Grading System in MSU Maguindanao", how will you make it a qualitative research?

References:

Books

- Almeida, A., et al (2016). *Research Fundamentals From Concept to Output. A Guide for Researchers & Thesis Writers.* Adriana Printing Co., Inc.

- Del Rosario-Garcia, M., et al (2017). *Practical Research 1 Basics of Qualitative Research*. Fastbooks Educational Supply Inc.

LESSON 8

DESIGN SCIENCE METHODS AND APPROACHES

Overview:

This lesson describes the technical terms, relevant algorithms, and possibly mathematics theorems for better understanding of the reader. For capstone projects that use or integrate existing software products, the latter should be described in sufficient detail.

Learning Outcomes:

At the end of this lesson, the students can:

1. write technical background of a given research topic; and
2. propose technologies to be used in developing and implementing a research project.

Materials Needed:

- Computer
- Projector
- Presentation Slide

Duration: 1 hour and 30 minutes

Learning Content:

The technical background may be long or short depending on the number of items that need to be explained and its complexity. This should be presented in simple and understandable form. There are no hard and fast rules on what could be placed in this section. Aside from texts, the author may put tables, graphs, illustration, pictures and other relevant information, as necessary.

Format

The technical background must be written in narrative form. It is important the chapter or section starts with a paragraph that describes what the readers should expect in it. Subheadings are recommended for descriptions that are substantially long. While there is no format, effort should be exerted by the writer to make specific items in the technical background accessible to the reader when the latter is needing references.

Considerations

It is the most likely that abbreviation and acronyms will be widely used in this section. The first time you see the abbreviation; you must write out the full form and put the abbreviation brackets. From then on you may use the abbreviation or acronym for the succeeding pages of the manuscript.

One should always follow the Philippine Copyright Law [12]. Proper citation should always indicate sources if the explanation are not from your own.

It is worth keeping in mind that this section may set the tone of the whole report or paper. Hence, it should be clear and understandable, possibly using different visual presentations as necessary.

Guidelines in Writing the Technical Background:

1. Overview of the current technologies (hardware/software/network) used in the current system
2. Discussions on the current trends and technologies to be used in developing and implementing the proposed system
    - HARDWARE
    - SOFTWARE
    - PEOPLEWARE
    - NETWORK
3. Fluidity and continuity should be observed

Learning Activities:

Activity 1:

Draw and discuss the current technologies used in the Grading System of MSU Maguindanao.

Activity 2:

Suggests and explain the technologies to be used in developing and implementing the proposed system, Grading System of MSU Maguindanao.

Learning Evaluation:

1. Write the Technical Background of your research topic.

References:

Sta. Romana, C. C, et al. (2012). *Undergraduate Research and Capstone Project Manual*. Philippine Society of Information Technology Educators, Inc

LESSON 9
PREPARING AND WRITING THE METHODOLOGY

Overview:

The design is a blueprint of the concept of the proposed research or capstone project. It provides an outline of the phases and sub phases that will help the proponents to be guided in their choice of the techniques that are most appropriate at each stage of the research or project. It will also help the project proponents plan, manage, control and evaluate computing research or projects

Learning Outcomes:

At the end of this lesson, the students can:

1. write a research methodology of a research topic;
2. analyze and design an appropriate system methodology for the development of a research project; and
3. formulate a schedule of timeline for the implementation of a research project.

Materials Needed:

- Computer
- Projector
- Presentation Slide

Duration: 1 hour and 30 minutes

Learning Content:

The methodology is defined as a collection of procedures, techniques, tools, and documentation aids which will help the project proponents in their efforts to solve a computing problems or implement a new information system. Hence, a methodology is a representation of how the research is to be solved or how the system proposed is to be developed. It should be supported with a theoretical basis, the philosophical views that led to the fundamental principles of the method, its interest, and the viewpoints or the people who invented the method.

Purpose

For undergraduate thesis or capstone project, the adviser places an important role on the choice of methods to use. The team of panelists that will review the thesis or project during latter’s proposal stage should provide necessary suggestions on possible ways to proceed given the proposed methods or critique the proposed methods and possibly suggest new once.

Considerations

The research design and methodology chapter highlight the technique and tools employed in the undergraduate thesis or capstone project.

Typical contents of this chapter and include research or project concepts, systems analysis and design approach, development model, development approaches, software development tools, scheduling and timeline of the research or project, project team and their responsibilities, budget proposal and cost management plan, and verification, validation and testing plans.

On implementation and Development

The implementation could mean a full implementation of system with software application development requirements. The system development consists of activities and processes aimed to developed, install and test proposed system. The development is more involved and often involves communication with system users and decision makers.

System Development

System development is a process that comprises a set of activities, methods, deliverables, and tools that student researches or developers may use to develop the proposed system. This process basically includes three major activities, and these are: analysis, design and implementation [6, 26].

System analysis focuses on system requirements that will determine and specified by the developers [18]. This process provides system requirements description; defines the system functional requirements, and requirements specification of the proposed system [6].

System provides the technical specification and construction of the solution for the requirements identified during the system analysis phase of the project/research [21]. It is consider as an essential part of the software engineering. It provides the details of data structure, architecture, Interface and procedural detail of software component of the research /project [21].

System implementation is the installation and the delivery of the proposed system to be conducted by the researchers/developers. It includes conversion and integration plan, database installation, system testing, user training and other production activities [6,25].

The documentation of other system development from system analysis to actual deployment provide the reader an understanding on how the system was analyzed, designed, and implemented. Student developers should provide a detailed documentation of each activity done during the analysis, design, and implemented phase. These presentations are in the form of tables, figures and other similar diagrams use in either structured or object- oriented approach. Therefore all student developers need to have an ideal understanding of the subjects" System Analysis and Design "and "software engineering".

Learning Activities:

Activity 1:

Analyze and Design an appropriate system methodology for the development of Grading System of MSU Maguindanao.

Activity 2:

Schedule a timeline for the implementation of the research project Grading System of MSU Maguindanao following a software development cycle.

Learning Evaluation:

1. Write the methodology of your research topic.

References:

Sta. Romana, C. C, et al. (2012). *Undergraduate Research and Capstone Project Manual*. Philippine Society of Information Technology Educators, Inc

LESSON 10
PRESENTING THE PROJECT PROPOSAL

Overview:

This lesson discusses guidelines in preparing a project proposal and its content.

Learning Outcomes:

At the end of this lesson, the students can:

1. present a research proposal

Materials Needed:

- Computer
- Projector
- Presentation Slide

Duration: 1 hour and 30 minutes

Learning Content:

ELEMENTS OF THE CAPSTONE PROPOSAL

The proposal is a statement of your intention to do this research project. It needs to convince the reader that you have: a) a clear idea and question; b) a thought-through methodology (way to solve your problem or answer your question); and c) a project that is doable within the scope of the degree requirements. You also need to demonstrate that you are capable and ready to do launch the main part of the project.

Here are the specific elements that need to be incorporated into the proposal in some way: Any of these elements can be separated out in designated sections or can be incorporated together in various areas of your text. You can write in a continuous narrative, create sections with headings, or use a combination, **as long as you are clear and the reader gets the following information from your proposal.**

Proposal Content Guidelines for Capstone Project

The proposal for the capstone project will consist of 10 parts given below. The Capstone Proposal needs to be in APA format 5th edition (review the APA anual for format features). References to support the project must also be cited in APA format and included under a References section. Use a Running Head in the proposal along with the page number. Capstone Proposals tend to be between 13-16 pages long.

•Abstract
•Introduction
•Goal(s) and Objectives
•Rationale and Needs Assessment
•System and Components
•Methodology of Implementation
•Work Plan
•Related Works and Literatures
•Signature block

Abstract: A 50 to 75 word abstract that discusses the main purpose of the project, its audience and educational value, medium(s) used, and its overall significance. The Abstract should be on a page of its own apart from the remaining content.

Introduction: The introduction will be a narrative description of the project including:
• A brief overview of the project's structure
• Compelling reason(s) why the project is being developed
•Who will be using the completed project
• The medium that the project will assume (CD, Web-based, etc.)

Goal(s) and Objectives: Give the overall goal and objectives of the project.

Goals are broad	Objectives are narrow.
Goals are general intentions;	Objectives are precise.
Goals are intangible;	Objectives are tangible.
Goals are abstract;	Objectives are concrete.
Goals can't be validated as is;	Objectives can be validated.

Rationale and Needs Assessment: Explain how you came about identifying the problem and what you would like to see happen. Discuss how your project will help your school/institution/organization as a whole.

System and Components: A narrative with detailed description of the system being pursued and its components. A bulleted list may be used to list the sub-components.

Methodology of Implementation:

Explain in detail how the system and components will be implemented. Identify the approach/ technique to be pursued. Explain the participation of each members and their learning intentions. This area shows the intellectual stretch of each members of the study.

Review of Related Works and Literature:

Contains the body of knowledge which serves as the foundations for building the Capstone Project. References to books, journals, and other materials are written here

Work Plan:

Gantt Chart with detailed tasks and timeline.

Learning Activities:

Activity 1:

Present your proposal on the research Grading System of MSU Maguindanao.

Learning Evaluation:

1. Present your research proposal.

References:

Barcenas, R., Teruel, A. (2012). *A Handbook for the Capstone Program of the BSIT Department.* https://www.slideshare.net/ledorsanecrab/capstone-primer-for-bsit-program-dbtc, Date Accessed Nov. 13, 2013